The paleomagnetism of the Great Slave Supergroup: the Akaitcho River Formation

M. E. Evans, G. S. Hoye, and D. K. Bingham

Institute of Earth and Planetary Physics, University of Alberta, Edmonton, Alta., Canada T6G 2J1

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Results are reported from 120 oriented samples collected from 40 stratigraphic horizons spanning the Akaitcho River Formation of the Great Slave Supergroup. Thermal demagnetization experiments reveal the presence of multiphase remanence. Difference vectors indicate that the lower blocking temperatures carry a magnetic component attributable to the present field. The remanence remaining above 500–600°C defines two polarity groups, but the two groups are not exactly antiparallel. This is interpreted in terms of a previously recognised widespread magnetic overprint acquired during uplift and cooling in the Coronation Geosyncline. A compilation of appropriate overprint data from the Great Slave Supergroup yields a mean direction of $D = 152°$, $I = +64°$ ($N = 75$ sites, $k = 30$, $\alpha_{95} = 3°$), with a corresponding pole at 91°W, 21°N ($K = 14$, $A_{95} = 5°$). A statistical procedure is employed to remove this overprint and obtain a best estimate of the original magnetization, which corresponds to a paleomagnetic pole on the western limb of the Coronation loop (92°W, 4°S, $N = 35$ sites, $K = 13$, $A_{95} = 7°$).

On rapporte les résultats de mesures effectuées sur 120 échantillons orientés prélevés à 40 niveaux stratigraphiques qui couvrent la formation d'Akaitcho River dans le supergroupe de Great Slave. Les expériences de désaimantation thermique révèlent la présence de rémanence multiphasée. Les vecteurs de différence indiquent que les températures inférieures de blocage portent une composante magnétique attribuable au champ actuel. La rémanence qui persiste au-dessus de 500–600°C définit deux groupes de polarité, mais les deux groupes ne sont pas exactement antiparallèles. On interprète cela comme une surimpression magnétique de grande envergure déjà reconnue et qui a été acquise durant le soulèvement et le refroidissement du géosynclinal de Coronation. Une compilation des données pertinentes de surimpression dans le supergroupe de Great Slave donne une direction moyenne de $D = 152°$, $I = +64°$ ($N = 75$ sites, $k = 30$, $\alpha_{95} = 3°$) avec un pôle correspondant à 91°O, 21°N ($K = 14$, $A_{95} = 5°$). On utilise un traitement statistique pour éliminer cette surimpression et obtenir une meilleure estimation de l'aimantation originale, laquelle correspond à un pôle paléomagnétique sur le flanc ouest de la boucle de Coronation (92°O, 4°S, $N = 35$ sites, $K = 13$, $A_{95} = 7°$).

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Introduction

Paleomagnetic investigations of Phanerozoic formations have provided much significant information concerning the Earth's structure and evolution, but the much longer Precambrian record is still in the early stages of investigation. For North America, important sources of data are the sedimentary basins of the Canadian Shield, which contain thick sequences of relatively unmetamorphosed and undisturbed sediments dating back to early Proterozoic times. This paper reports results from the Athapuscow Aulacogen — an elongated basin closely associated with the Coronation Geosyncline (Hoffman *et al.* 1970; Hoffman 1973). Samples were collected from the Akaitcho River Formation which represents a small fraction of the Great Slave Supergroup — a sequence of some 8000 m of sedimentary rocks deposited between about 2200 and 1800 Ma ago. The Great Slave Supergroup is divided into four lithological groups, each of which is further subdivided into formations; the Akaitcho River Formation is the youngest of four formations comprising the oldest group (the Sosan Group). Its age is well-established because it is laterally equivalent to the Seton Formation, a volcanic sequence which has been dated radiometrically by Baadsgaard *et al.* (1973) using the Rb–Sr method, and by Cumming (in press) using lead isotopes. Cumming also quotes unpublished results by the Geological Survey of Canada, and recalculates the ages using the decay constants recommended by Steiger and Jager (1977) ($\lambda^{87}Rb = 1.42 \times 10^{-11}$/year, $\lambda^{238}U = 1.55125 \times 10^{-10}$/year, $\lambda^{235}U = 9.8485 \times 10^{-10}$/year). He obtains Rb–Sr ages of 1833 ± 28 Ma and 1805 ± 16 Ma, whist his own Pb/Pb work yields an age of 1804 ± 23 Ma. The agreement between the methods is excellent and thus the date obtained probably corresponds to the initial formation of the Seton lavas.

Field and Laboratory Methods

The Akaitcho River Formation consists of red micaceous siltstones and red shales reaching a maximum thickness of about 300 m, but no single

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section spanning the entire Formation could be located. Instead, samples were collected from three sections (B, C, and D, Fig. 1), each of which is some 200 m thick. The sampling horizons could not be securely tied to the top of the formation, but in the C section a sharp contact with the underlying Kluziai Formation is clearly visible. Also, the lowest site in the D section is located within a thin reddish-pink layer underlain and overlain by white quartzites, and is actually in the uppermost Kluziai. At each sampling site three (occasionally two or four) oriented drill cores or hand blocks were obtained with as little stratigraphic spread as possible. In one or two cases the nature of the outcrop required that this be as much as a metre, but in most cases a sampling site involved no more than 20 cm stratigraphically. The B, C, and D sections consist of 21, 19, and 10 sites respectively. A total of 120 separately oriented samples was collected from the 40 sites (102 drill cores and 18 hand blocks). The majority (108) were oriented by solar bearings, but in the remaining 12 cases orientation was magnetic. The 108 opportunities for cross-checking the two methods demonstrate excellent agreement, with an average difference of 0.5 ± 2.2° (standard deviation). All measurements of remanence were made on a PAR SM-1 spinner magnetometer interfaced to a Wang 600 desk-top computer; thermal demagnetization employed a small oven in a room-sized magnetic shield (Reid 1972), and alternating-field (AF) demagnatizations were carried out using the apparatus described by Murthy (1969). One specimen from each of 20 sites was selected for pilot thermal demagnetization studies; 10 were heated to temperatures of 100, 200, 300, 400, 450, 500, 550, 575, 600, 630, 660, and 700°C, and 10 to temperatures of 400, 450, 500, 550, 575, 600, 630, 660, and 700°C. One specimen from each of the remaining samples was subjected to an abbreviated heating sequence between 400 and 600°C, the exact temperatures depending on behaviour.

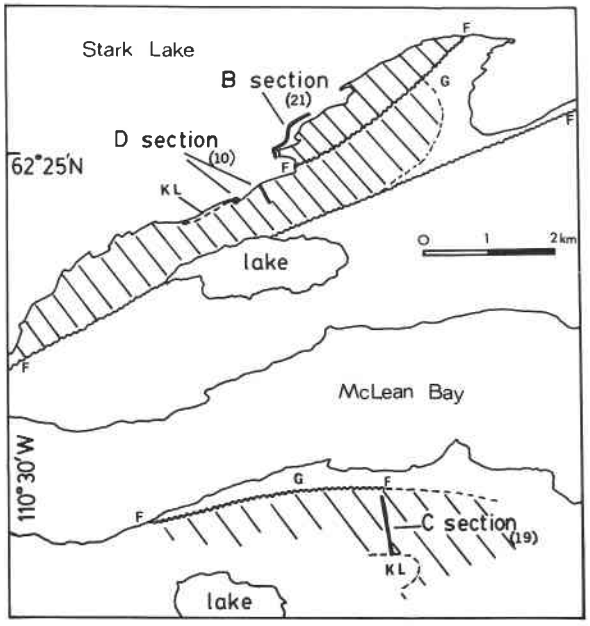


FIG. 1. Geological sketch map showing the outcrop of the Akaitcho River Formation (oblique lines) in the McLean Bay area of the East Arm of Great Slave Lake, N.W.T. The stratigraphic sections sampled (B, C, D) are shown and the number of sites in each are given in parentheses. KL, Kluziai Formation (which underlies the Akaitcho River Formation); G, Gibraltar Formation (which overlies the Akaitcho River Formation); F, fault.

Results

The specimens are quite strongly magnetized, with natural remanent magnetizations (NRM) ranging from 0.7×10^{-3} to 19×10^{-3} A/m, and a median value of 3×10^{-3} A/m. During heating there is a strong tendency for remanence vectors to move away from the present field at the sampling locality ($D = 030°$, $I = +83°$), suggesting the removal of a recently acquired secondary component of magnetization. This is confirmed by analysis of the magnetic vector removed by heating to 500°C, which is the lowest temperature to which all specimens were heated. The magnitude of the remanence removed in this step ranges from 16–240% of the initial (NRM) intensity with a median of 68%. These difference vectors are first averaged at each site and the lower quality data eliminated by means of Watson's (1956) test for randomness. This leaves 33 sites with an average of $D = 075°$, $I = +83°$ ($k = 31$, $\alpha_{95} = 4.5°$), which is very close to the present field at the sampling locality ($D = 030°$, $I = +83°$). After correcting for geological dip of the beds the precision is reduced to 5.3, which is a significant decrease at a confidence level exceeding 99% (McElhinny 1964), indicating that folding predates the secondary magnetization, as expected.

Although these results strongly suggest the presence of a recent overprint, closer scrutiny reveals a slight offset which is probably significant. The circle of confidence has a radius of 4.5°, but the angle between the present field and the difference vector mean is 5.4°. The angular divergence from the axial dipole direction, which is perhaps a more appropriate reference, is 14.5°. This offset is towards an earlier, geologically ancient, overprint direction which is discussed in detail below. The displacement is presumably due to the fact that heating to 500°C removes all of the recent overprint and also some of the earlier overprint. Detailed thermal de-

magnetization below 500°C might yield a cleaner separation of these two overprints, depending on their blocking temperature spectra, but this has not been attempted in the present study.

In the lower temperature ranges large angular shifts take place often accompanied by substantial changes in intensity, but at higher temperatures directions generally stabilize. Some specimens remain stable beyond 600°C but others become erratic, often undergoing marked intensity increases suggesting that irreversible chemical changes have occurred. Between 500 and 550°C angular shifts are generally small (69% ≤ 5°:90% ≤ 10°), with only four shifts greater than 16° being observed. Of these four, two agree with companion samples at 500°C but are not stable up to 550°C, and the other two do not reach end points until 550°C but exhibit only small changes thereafter (8 and 9° between 550 and 575°). These observations suggest that meaningful remanence directions have been isolated from the great majority of samples.

We follow the standard practice of grouping the data to obtain site mean results—each mean representing a "spot" reading of the ancient geomagnetic field. In most cases the 500 or 550°C treatments were used, the one with the higher precision parameter (k) being accepted. In a few cases higher treatments were favoured since some of the specimens involved were still undergoing small systematic movements. The bulk of the sites respond to thermal cleaning in a straightforward manner and yield two populations of opposing polarities (Fig. 2, Table 1), but six sites (BA, CD, CF, DB, DD, and DF) behave differently. Specimens from sites BA and CD move systematically away from the present geomagnetic field towards the northerly group obtained from many of the other sites, but never quite attain that goal, suggesting that the secondary overprint is such that the underlying remanence can be almost, but not quite, isolated by thermal treatment. Alternating-field demagnetization and chemical leaching were tried but neither technique proved effective and no additional information was obtained. Sites CF and DB yield statistically significant means removed from the present geomagnetic field, but divergent from the overall mean direction indicated by the bulk of the data; site DD is similarly divergent but apparently contains two opposed polarities. Site DF yields well-grouped NRM directions near the southerly group (Fig. 3), but on demagnetization the remanence vectors migrate towards a northerly direction in a manner consistent with the selective removal of a component of one polarity in the presence of an opposed polarity.

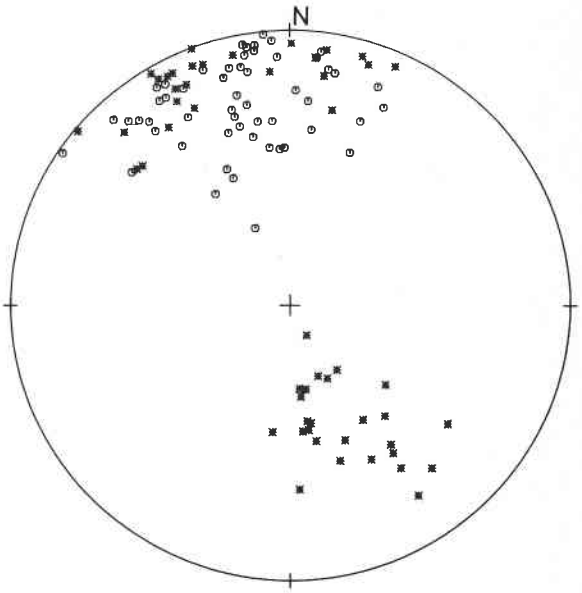


FIG. 2. Thermal demagnetization results. The points represent sample directions (N = 106) at the temperatures given in Table 1 (parenthesized entries are not plotted). Equal area projection, open symbols on upper hemisphere.

Discussion

Magnetostratigraphy

Sections C and D contain magnetizations of both polarity with a transition from normal to reversed taking place about 100 m above the base of the formation (Fig. 4). The B section contains no polarity transitions, and since the entire Akaitcho River Formation is reported to be not much thicker than 300 m, the combined thickness of the sections implies that the B section reaches close to the top of the formation.

The polarity reversal found in the C and D sections is a potentially useful stratigraphic marker. Obviously it allows local correlation, and work currently in progress is aimed at establishing whether it can be used to confirm lithological correlations over several hundred kilometres between different parts of the Coronation Geosyncline.

A reversed polarity group is clearly defined by the B section, but the upper parts of the C and D sections are more complex. Site CI indicates a return to normal polarity at the top of the C section; this may be an "event" but its duration cannot be determined since site CI is from an isolated outcrop offering no opportunity of extensive sampling. Furthermore, the location of the site in thickly-wooded terrain and the proximity of a fault renders the actual stratigraphic position rather uncertain. Sites CF, DB, DD, and DF were mentioned above in connection with their behaviour during thermal

TABLE 1. Site mean results (dip corrected)

Site	N	D (°)	I (°)	k	T (°C)
BA	3	(345	+27	79	660)
BB	3	338	−35	15	550
BC	3	331	+18	137	500
BD	3	347	−46	42	550
BE	3	355	−02	96	500
BF	3	014	−06	7	550
BG	3	325	−15	57	500
BH	3	334	+06	114	550
BI	3	358	−20	28	550
BJ	3	356	−24	18	500
BK	3	339	−21	12	550
BL	3	320	+26	47	500
BM	4	348	−33	11	550
BN	3	358	+10	29	500
BO	3	331	−12	400	550
BP	3	349	−02	56	500
BQ	3	327	−39	12	500
BR	3	318	−20	87	500
BS	3	310	+05	57	500
BT	3	343	−09	7	550
BU	4	348	−15	163	550
CA	3	172	+53	1630	550
CB	3	164	+56	44	500
CC	3	149	+47	156	550
CD	3	(005	+08	8	660)
CE	3	022	−03	23	600
CF	3	(291	+86	17	500)
CG	3	335	+02	57	600
CH	3	003	+05	66	550
CI	3	172	+64	2400	500
DA	3	003	−16	62	550
DB	2	(100	−52	131	500)
DC	3	004	−09	11	630
DD	3	(	random		)
DE	3	359	−25	16	600
DF	3	158	+34	33	0
DG	2	142	+29	77	500
DH	3	145	+30	20	550
DI	3	162	+54	16	500
DJ	3	142	+67	31	500

demagnetization. Before thermal treatment, site DF exhibits normal polarity, but as demagnetization proceeds a reversed component is also indicated (Fig. 3), which is consistent with the location of this horizon between sites DG (normal) and DE (reversed). Sites CF, DB, and DD do not occupy "transitional" stratigraphic positions — this does not preclude them from representing transitions to unrecognised polarity zones, but they may equally well represent geomagnetic excursions, or sites where thermal demagnetization fails to resolve multicomponent magnetizations. It is interesting to note that two specimens from site DD yield magnetic vectors similar to those found at site CF (Fig. 3), and since the stratigraphic positions of the two sites are comparable (Fig. 4) it is possible that they record the same geomagnetic feature. A somewhat

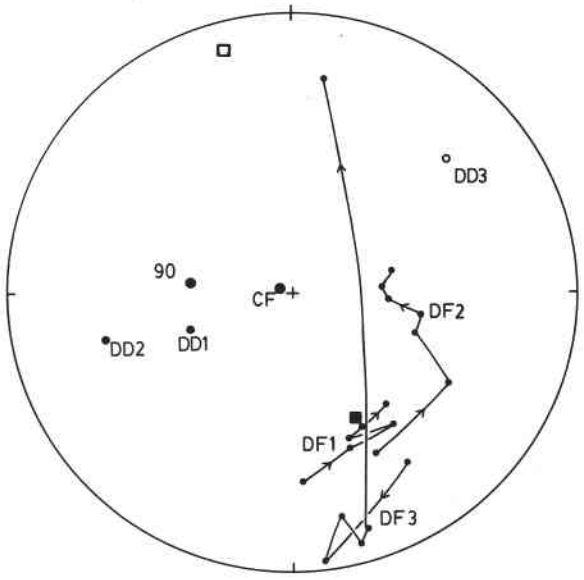


FIG. 3. Thermal demagnetization behavior at specific "nonstandard" sites. Equal area projection, open symbols on upper hemisphere. Site 90 of Irving and McGlynn (1979) is indicated. Squares represent the means of the two polarity groups after correction for geological tilt (see Table 2). Temperatures as follows: DD1, DD2, DD3 (600°C):DF1, DF3 (20, 500, 550, 600, 630, 660°C), DF2 (20,400, 500, 550, 600, 630, 660°C).

less certain correlation is suggested between sites CI and DB (Fig. 4), but the uncertainty in the stratigraphic position of CI and the rather aberrant direction at site DB (Table 1) renders this rather speculative. Indeed, the sampling density is such that all the suggested correlations should be regarded as rather tentative at this stage. If these features represent real geomagnetic phenomena their absence from the B section means either that they fall between sampling sites or that the B section lies completely above sections C and D, with the Akaitcho River Formation, consequently, totalling some 400 stratigraphic metres.

Site Mean Directions

We have seen that the bulk of the data (35 out of 40 sites) falls into two populations of opposing polarity (Fig. 2). However, the means of the two groups are only 142 ± 14° apart, not 180°, as would be expected. This situation is often indicative of the presence of secondary components of magnetization, and, therefore, demands careful scrutiny. When the two groups are analyzed separately (Table 2), both show improved precision with the strata restored to the horizontal, but only for the northerly (reversed) group is the improvement significant at the 95% level (McElhinny 1964). The combined group (obtained by inverting one of the

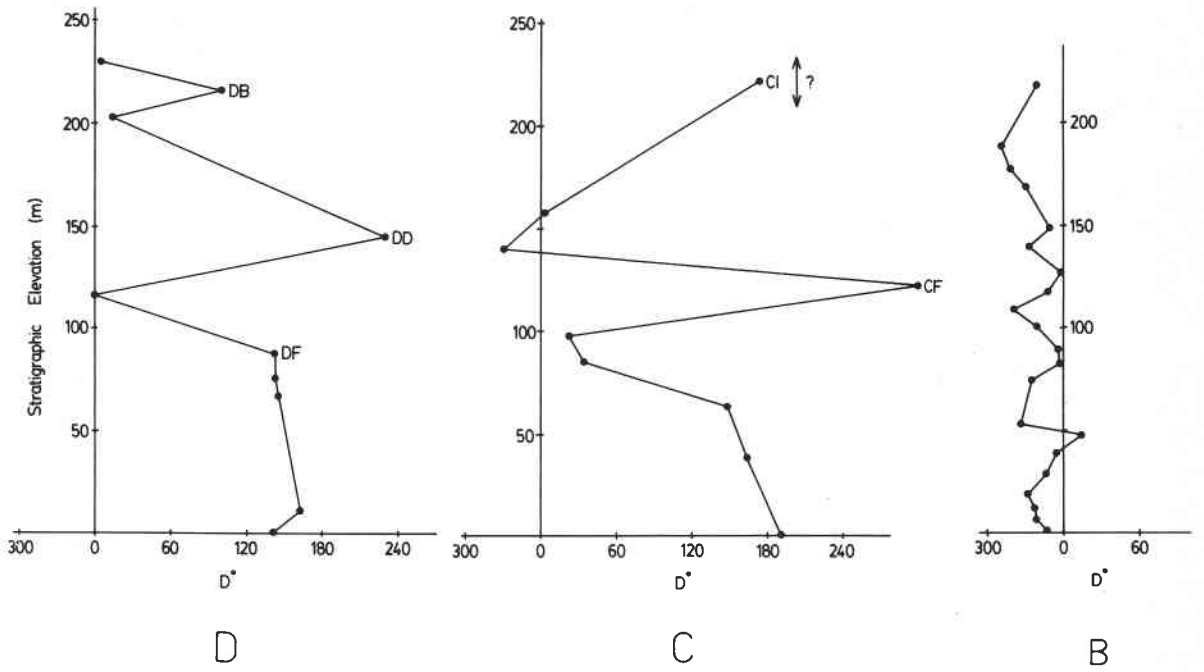


FIG. 4. Declination magnetograms for sections B, C, and D. Possible stratigraphic correlations between C and D are discussed in the text.

polarity groups) passes the fold test at the 99% level (Table 2). Thus, it appears that a substantial fraction of the magnetization was acquired before the folding took place, i.e., before deposition of the Et-Then Group, which unconformably overlies the Great Slave Supergroup. We return to a fuller discussion of possible magnetic overprints in the next section.

There are two ways of approaching the observed asymmetry. It can be attributed to a magnetic overprint, in which case the usual procedure is to calculate the mean irrespective of sign. This gives $D = 162°$, $I = +21°$ ($N = 35$, $k = 9$, $\alpha_{95} = 9°$). Alternatively, it can be assumed real, implying either that the geomagnetic field actually was asymmetric, or that some polar wander took place as one polarity epoch succeeded the other. In either case, it is appropriate to report a separate direction and pole for each group (Table 2). A similar pattern has been reported from other Great Slave Supergroup formations by Reid (1972) and Irving and McGlynn (1979). Reid's work (Reid 1972; McMurry *et al.* 1973) involves a few Seton Formation sites but mostly concerns the overlying Kahochella Group. Irving and McGlynn investigated tuffs and tuffaceous siltstones of the Seton Formation, which is laterally equivalent to the Akaitcho River Formation. Both polarities are present in their data and the same asymmetry is observed. The only significant difference is that most of their sites yield southerly declinations (normal), suggesting that the Seton Formation (or their sampling of it) is, for the most part, equivalent to the lower parts of our sections C and D. Their sampling may, of course, extend to lower horizons. A further point of interest in connection with the Seton results is the occurrence of an aberrant direction at site 90, which is similar to our results for sites CF and DD (Fig. 3), and lends credence to the suggestion that a real geomagnetic feature is involved.

TABLE 2. Summary of results from reversed and normal polarity groups

		Directions			Poles	
Group	N	D (°)	I (°)	k	Lat.	Long.
Reversed	26	346	+14	5.6	—	—
Reversed	26	345	−11	11.1	21°S	094°W
Normal	9	161	+62	13.0	—	—
Normal	9	155	+49	25.1	04°N	089°W
Combined	35	345	−05	3.4	—	—
Combined	35	342	−21	8.6	16°S	094°W

NOTE: The lower entry in each pair is the dip corrected one.

The Coronation Overprint

Magnetic overprints have been reported from all groups of the Great Slave Supergroup (McMurry *et al.* 1973; Reid 1972; Evans and Bingham 1976;

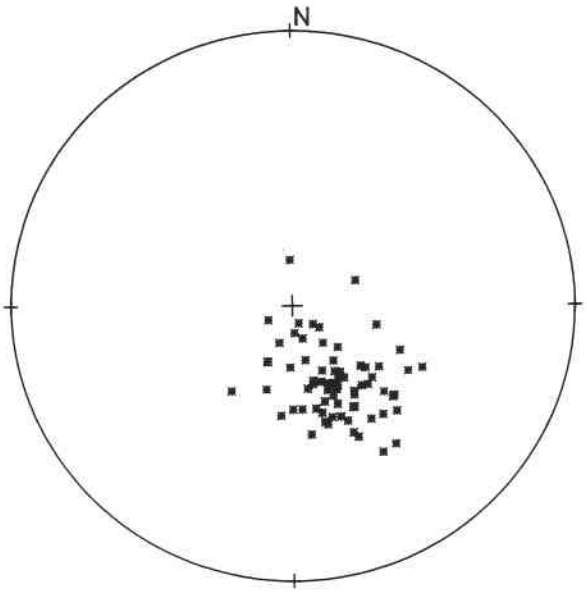


FIG. 5. Summary of magnetic overprint vectors obtained from Great Slave Supergroup Formations. All points on lower hemisphere of equal area projection. $N = 75$ (Seton, 14; Kahochella, 37; Pearson, 7; Akaitcho River, 4; Tochatwi, 13).

Bingham and Evans 1976; McGlynn and Irving 1978; Irving and McGlynn 1979; Reid *et al.* 1980), and it is natural to enquire if these could be responsible for the lack of antiparallelism in the Akaitcho River Formation results. There are two identifiable overprints, one due to reheating by the Mackenzie intrusions (−1250 Ma), and one conjectured by McGlynn *et al.* (1974) to be related to uplift and cooling following orogeny in the Coronation Geosyncline (about −1700 Ma, see also McGlynn and Irving (1978)). The former is not a feasible source of the discrepancy in question as its direction is quite inappropriate, lying, as it does, almost perpendicular to the plane defined by the two polarity groups in the Akaitcho River Formation. The latter, however, constitutes a viable possibility, which we now assess.

Figure 5 summarizes magnetic overprint data from several formations in the Great Slave Supergroup. The 75 collecting sites involved span several hundred stratigraphic metres and are spread over some 80 km along the East Arm of Great Slave Lake. Their mean direction is $D = 152°$, $I = +64°$ ($N = 75$ sites, $k = 30$, $\alpha_{95} = 3°$). We enquire if such an overprint could be responsible for the observed reversal asymmetry in the following way. From the "cleaned" magnetization vector of each specimen (corresponding to the site entries in Table 1) an overprint vector in the direction 152, +64 was removed, with the strata in their present attitudes. The magnitude of this overprint was set at a particular fraction of the "cleaned" remanence. Various statistics were calculated and the magnitude of the overprint vector was then incremented and the whole procedure repeated. The method is very

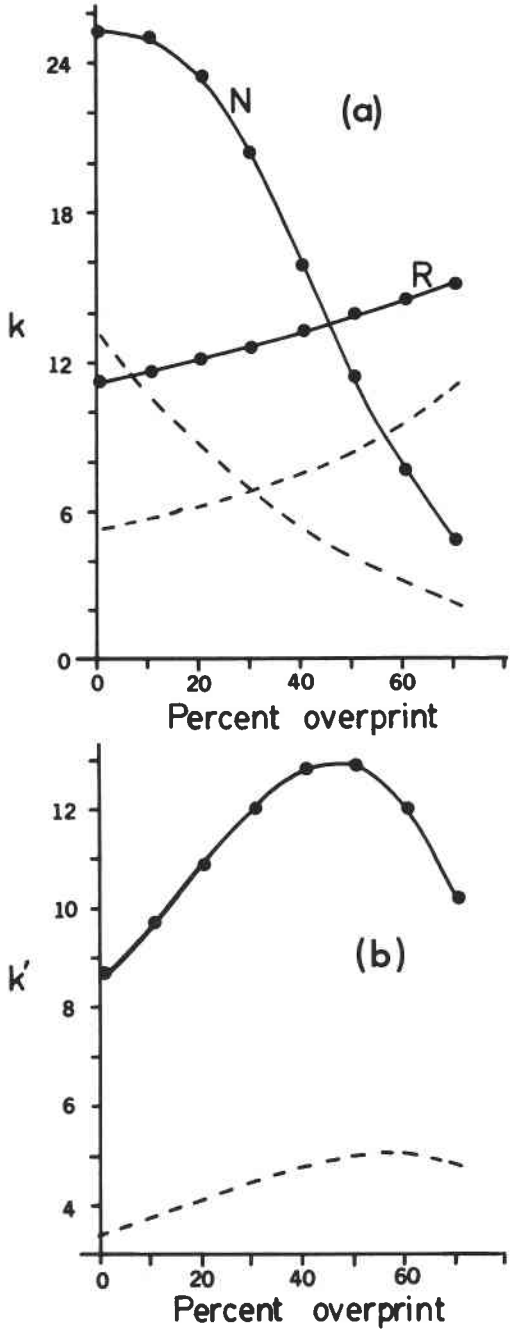


FIG. 6. (*a*) Fisher's precision parameter (k) for the two polarity groups of Fig. 2 as a function of overprint removed (see text). The normal group is labelled N, and the reversed R. Solid (dashed) curves are after (before) application of bedding correction. (*b*) Precision parameter (k') for the combined population (N + R) as a function of overprint removed (see text).

similar to that described by Williamson and Robertson (1976). The outcome is that the two polarity groups, after application of the bedding correction, gradually move towards antiparallelism, with concomitant changes in their internal variances (Fig. 6*a*). The precision parameter (k') of the combined population (obtained by inverting one of the groups in the usual manner) peaks when the overprint is 45% of the "cleaned" remanence (Fig. 6*b*), and it is noteworthy that this is the level at which the internal variances of the two polarity groups become equal. We regard this as the optimum trade-off between exact antiparallelism and the internal scatter of the two groups. At this point the overall precision ($k' = 13$) is typical of results from comparable formations (Stark: $N = 39$, $k = 13$; Tochatwi: $N = 17$, $k = 14$; Douglas Peninsula, high temperature: $N = 6$, $k = 14$; Seton "C": $N = 19$, $k = 24$; Takiyuak: $N = 17$, $k = 15$; Et-Then: $N = 14$, $k = 17$; see Evans and Bingham (1976), Bingham and Evans (1976), and Irving and McGlynn (1979)). Furthermore, both polarity groups now pass the fold test (reversed group: before dip correction, $k = 7.9$, after, $k = 13.4$; normal group: before dip correction, $k = 4.6$, after, $k = 13.5$) and their means are antiparallel within experimental uncertainty, their angular separation being $166 \pm 17°$. Finally, it is noted that the combined group, after removal of the overprint, passes the fold test at the 99% level ($k' = 13$ compared to $k' = 5$).

Paleomagnetic Poles

The Coronation overprint yields a paleomagnetic pole at 91°W, 21°N ($N = 75$ sites, $K = 14$, $A_{95} = 5°$), corresponding to uplift and cooling at about −1700 Ma (McGlynn *et al.* 1974). The uncontaminated remanence direction revealed by the statistical procedure described above has a mean direction of $D = 160°$, $I = +36°$ ($N = 35$, $k = 13$, $\alpha_{95} = 7°$), which we believe is the most reliable estimate of the original magnetization of the Akaitcho River Formation, with a corresponding age of about −1800 Ma. It corresponds to a paleomagnetic pole on the descending limb of the so-called Coronation loop at 92°W, 4°S ($N = 35$ sites, $K = 13$, $A_{95} = 7°$). This is about 7° south of that reported by Irving and McGlynn (1979) for the original magnetization of the Seton Formation. This is not a significant difference, but we suggest that it is real and is due to the presence in the Seton Formation data of a "Coronation" overprint as reported here for the Akaitcho River Formation.

Acknowledgments

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